

Probabilistic Circuit Learning and Spectral Tensor Decomposition Benchmarks

1 Introduction

This report details the computational benchmarks and verification metrics for learning Probabilistic Circuits (PCs) via the STRUDEL algorithm, followed by continuous parameter extraction using a customized Spectral Tensor Decomposition framework. The pipeline processes structural data, computes circuit flows via Maximum Likelihood Estimation (MLE), and extracts true marginal bag weights (μ) and coin biases (α) using polynomial identity testing and orthogonal matrix whitening, tensor decomposition.

Benchmarks were executed on a local environment across standard discrete datasets from the UCLA StarAI suite.

2 Execution Running Times & Verification Metrics

Table 1 summarizes the running times for the algorithmic portion of STRUDEL alongside the Tensor Extraction processing times.

To verify mathematical exactness, 100 queries from the actual test datasets were evaluated. The table reports the Maximum Absolute Error across these 100 queries between the black-box STRUDEL Probabilistic Circuit output and the recovered analytic mathematical formula.

Table 1: Running Times and Extraction Accuracy across StarAI Discrete Datasets

Dataset	Variables	Train Samples	Strudel Time (s)	Tensor Extraction Time (s)	Max Absolute Error
NLTCS	16	16,181	0.7254	0.1304	1.53×10^{-13}
MSNBC	17	291,326	13.5010	0.1209	2.85×10^{-13}
PLANTS	69	17,412	3.3186	0.9823	1.26×10^{-18}
BAUDIO	100	15,000	4.0543	2.0338	5.37×10^{-20}
JESTER	100	9,000	2.7734	2.0357	1.39×10^{-24}
BNETFLIX	100	15,000	4.2512	1.9161	9.28×10^{-31}
ACCIDENTS	111	12,758	3.9000	2.4461	1.88×10^{-24}
CONNECT4	126	16,000	5.8218	3.5889	3.30×10^{-19}
PUMSB_STAR	163	12,262	5.2438	5.0410	8.71×10^{-28}
DNA	180	1,600	0.7550	6.4108	9.37×10^{-48}
CWEBKB	839	2,803	7.0300	154.2184	1.32×10^{-36}
CR52	889	6,532	17.3681	178.9978	8.57×10^{-28}
VOTING	1,359	1,214	4.8871	418.1562	8.47×10^{-153}
MOVIEREVIEW	1001	1,600	4.9530	208.0012	2.27×10^{-104}
MUSHROOMS	112	2,000	0.6484	2.8460	4.88×10^{-12}
NIPS	500	400	0.6398	59.6366	8.62×10^{-110}
OCR_LETTERS	128	32,152	11.5730	3.6195	7.51×10^{-29}
RCV1	150	40,000	17.3512	5.2719	2.89×10^{-22}